Microeconomic Theory: TA Session

Pinda Wang

Johns Hopkins University

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Next week

Fall Break: No TA section next Friday Oct 23

Office hour: My office hour next week will be **on Zoom**, Wednesday Oct 21, 14:30-16:30 (1 hour longer than usual)

- Zoom link will be provided via a Canvas announcement
- However, if you want to attend next week, you'll need to email me (pwang66@jhu.edu) at least 1 day in advance

Midterm exam

You MAY take the midterm in class on Mon 10/26 or Wed 10/28

- So you choose ONE, and only one, of those class times to take the midterm
- You do NOT need to inform me or Prof. Daley of which date you will choose
- (Obviously, you do not need to come to class on the date your are NOT taking the midterm)
- No TA office hour on Wednesday 10/28; no TA section on Friday 10/30

Midterm coverage will be cumulative through class on 10/14

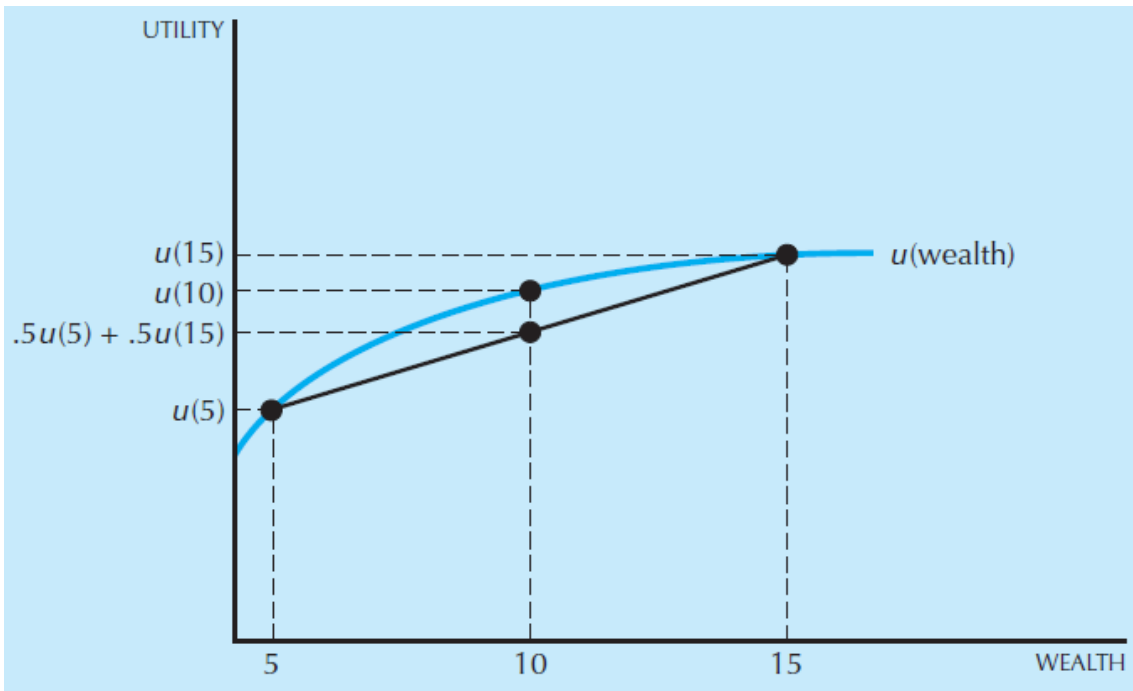
Risk attitudes and utility functions

In this graph, $u''(x) < 0$: if an individual's utility function is concave, then he's risk-averse

Notice that $u''(x) < 0$ also means that he has diminishing marginal utility

These two properties are connected! Both come from $u''(x) < 0$.

- For a person with diminishing marginal utility, losing \$5 is more painful than gaining \$5 is beneficial, so he's risk-averse



Risk attitudes and utility functions

For an increasing utility function ($u'(x) > 0$):

1. $u''(x) < 0$ (concave): risk-averse
2. $u''(x) = 0$ (linear): risk-neutral
3. $u''(x) > 0$ (convex): risk-loving

Homework 7.3

The Orioles are about to play in the seventh and deciding game of the World Series. The Horseshoe Casino is giving even odds that they will win: so if you bet \$1 and they win, you gross \$2; but if they lose you are out your dollar. You, however, noticed something about the opposing pitcher that leads you to think that the Orioles have a 0.75 probability of winning the game. You are also an expected utility maximizer with $u(I) = \sqrt{I}$ and with current total wealth of \$500.

(a) You are considering placing a \$15 bet.

(i) How much total wealth would you end with if the bet pays off? How about if you lose?

Answer: \$515; \$485

(ii) Should you make the bet (assume you have no principled opposition to gambling)? Explain.

Answer: EU from betting is $0.75\sqrt{515} + 0.25\sqrt{485} = 22.526$
EU from not betting is $\sqrt{500} = 22.361$
So, you should make the bet.

Homework 7.3

The Orioles are about to play in the seventh and deciding game of the World Series. The Horseshoe Casino is giving even odds that they will win: so if you bet \$1 and they win, you gross \$2; but if they lose you are out your dollar. You, however, noticed something about the opposing pitcher that leads you to think that the Orioles have a 0.75 probability of winning the game. You are also an expected utility maximizer with $u(I) = \sqrt{I}$ and with current total wealth of \$500.

(b) Now you are considering betting any amount between 0 and 500 (betting any fraction of a dollar is allowed). What dollar bet amount maximizes your expected utility? Justify your answer.

Answer: Suppose you bet x dollars.

$EU = 0.75\sqrt{500 + x} + 0.25\sqrt{500 - x}$, we want to maximize EU.

Taking derivative: $\frac{dEU}{dx} = \frac{0.75}{2\sqrt{500+x}} - \frac{0.25}{2\sqrt{500-x}} = 0$

Rearranging: $\sqrt{500 + x} = 3\sqrt{500 - x}$

Solving, we get $x = 400$.

Insurance - exercise

Jessica have \$10,000. There is a 10% probability that a disaster happens and she loses \$9,000. Suppose she can buy x dollars of insurance at a price of γ per dollar insured.

1. What is the actuarily fair value of γ that makes the insurance company earn zero profit?

   Answer: The insurance company's profit is
   $\gamma x - 0.1x = 0 \;\Rightarrow\; \gamma = 0.1$

2. Suppose Jessica's utility of wealth is $u(w) = \ln w$. Derive the optimum amount of insurance that she buys.

   Answer: Suppose she buys x dollars of insurance. Her expected utility is $EU = 0.1 \ln(1000 + 0.9x) + 0.9 \ln(10000 - 0.1x)$.
   Taking derivative with regard to x: $\frac{dEU}{dx} = \frac{0.09}{1000+0.9x} - \frac{0.09}{10000-0.1x} = 0$
   (use the chain rule: if $h(x) = f(g(x))$, then $h'(x) = f'(g(x))g'(x)$.)
   Rearranging, we get $1000 + 0.9x = 10000 - 0.1x \;\Rightarrow\; x = 9000$.
   She'll be fully insured.

Insurance - exercise

Jessica have \$10,000. There is a 10% probability that a disaster happens and she loses \$9,000. Suppose she can buy x dollars of insurance at a price of γ per dollar insured.

3. Suppose we don't know the exact form of $u(w)$; all we know is that Jessica is risk averse. Does your answer to Question (2) change?

   Answer: Her expected utility is
   $EU = 0.1u(1000 + 0.9x) + 0.9u(10000 - 0.1x)$.

   Taking derivative: $0.09u'(1000 + 0.9x) - 0.09u'(10000 - 0.1x) = 0 \;\Rightarrow\; u'(1000 + 0.9x) = u'(10000 - 0.1x)$.

   Since $u''(w) < 0$, it must be that $u(1000 + 0.9x) = u(10000 - 0.1x) \;\Rightarrow 1000 + 0.9x = 10000 - 0.1x \;\Rightarrow x = 9000$.

   It turns out that a risk-averse individual always prefers to be fully insured, regardless of the utility function form.

Insurance - exercise

Jessica have \$10,000. There is a 10% probability that a disaster happens and she loses \$9,000. Suppose she can buy x dollars of insurance at a price of γ per dollar insured.

4. For this question, Jessica's utility function is again $u(w) = \ln w$. Now suppose that, instead of the actuarily fair value, the insurance company wants to make a profit and charges $\gamma = 0.2$. Does your answer to Question (2) change?

   Answer: Now, $EU = 0.1 \ln(1000 + 0.8x) + 0.9 \ln(10000 - 0.2x)$.

   Taking derivative: $\frac{dEU}{dx} = \frac{0.08}{1000+0.8x} - \frac{0.18}{10000-0.2x} = 0 \Rightarrow$
   $0.08(10000 - 0.2x) = 0.18(1000 + 0.8x) \Rightarrow x = 3875$.

   If the insurance premium is higher than the fair value, the risk-averse individual will prefer to be partially insured.